HOW THE BIG DATA WORLD IS AFFECTING GEOSCIENCE EDUCATION: A STUDY IN HIGH-RANKED U.S. COLLEGES

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Abstract

The shift of our world from a service-based, industrial society to a knowledge-based, big-data world has proven to touch the lives of people worldwide. Geoinformatics is an emerging field that facilitates this transition within the earth-science community. Geoinformatics refers to the science and technology used to develop information science infrastructure to address problems and interpretations of large-scale geological phenomena. The benefits of geoinformatics are constantly growing as scientific and technological advances in Artificial Intelligence and large data sets queries continue to develop exponentially, yet the education surrounding geoinformatics fails to retain the same pace. A manual search for geoinformatics courses in the course catalogs of U.S. colleges within the best geology departments ranked by U.S. News revealed that these top programs have little to no courses involving geoinformatics. However, in ranking the schools in order of geoinformatics courses offered within their geology courses, it was found that the Western colleges provided more geoinformatics courses than the Eastern colleges with few exceptions. This phenomenon can be understood by looking at the differences in history, cultural attitudes, and values of the West in comparison to the East. For example, Silicon Valley, a region that serves as a global center for advanced technology and innovation, along with several other big data headquarters, such as Google, Facebook, Linkdn, are located in the Western U.S. Although education surrounding geoinformatics is sparse, the literature review finds that knowledge in the form of geoinformatics is of vital importance and provides benefits in several geo-applications and disciplines. Thus, the U.S. education system should shift focus towards geoinformatics gaining prominence within the school earth-science education in the U.S. The implications of this research prompt an urgent need for a detailed scholarly analysis of research in this area and curation of a standardized curriculum for geoinformatics within the U.S. college system. The permeation of geoinformatics through these methods would properly equip our future geologists with the knowledge required for the big data world needs of 2050.

Introduction

A Google Scholar search of the phrase "Geoinformatics" yielded 181,000 results, while a search for "Geoinformatics Education" produced 28,400 hits, primarily from Europe. The literature review revealed that studies on geoinformatics education are mainly located in Europe, with few originating from the United States. Blišťan's study (Blišťan et al., 2015) highlights the importance of geoinformatics education, emphasizing its role in contributing to the development of a knowledge-based society. Despite the United States' reputation as a technological leader, the literature suggests a lack of geoinformatics education in the country.

This gap prompted the hypothesis that U.S. universities may not prioritize geoinformatics courses. Amongst the literature review, there was no established or explicit definition for "geoinformatics". So, it is important to acknowledge this "grey zone" because it properly addresses the complexity and ambiguity of geoinformatics and highlights the need for careful consideration and critical thinking to interpret this study.

Methods

- Utilized the 2022 "Best Geology Programs" rankings from "U.S. News & World Report" to identify colleges to source data from
- Manual search (& email) conducted for undergraduate geoinformatics courses within the geology courses offered at these colleges using the department's course catalog on the school's websites
- Five colleges were excluded due to the unavailability of general course catalogs
- The literature review revealed a lack of an established or explicit definition for "geoinformatics"
- To identify geoinformatics courses, specific keywords and branches of geoinformatics were employed
- Loose, general definition of geoinformatics: the science and technology of handling, analyzing, interpreting , and communicating geospatial data using computer-based tools and techniques

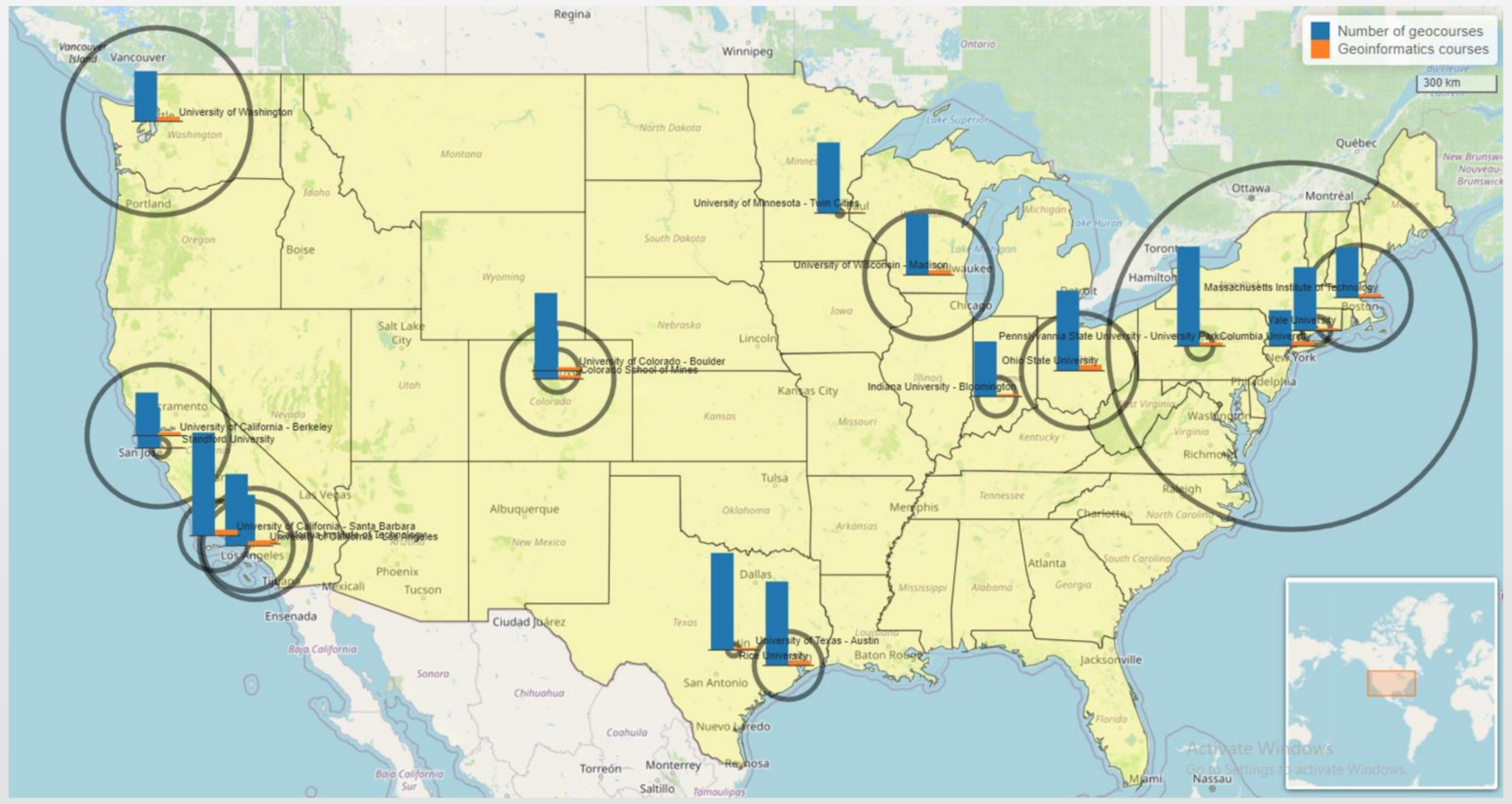

Figure 1: GIS Map of top-ranked U.S. Colleges in study (created in R). The radius of black rings reflect the percentage of geoinformatics courses per high-ranked US college.

Table 1: Results for each college including the number of geology courses offered, the number of identified geoinformatics courses within those offerings, and their percentage of geoinformatics course to total geology courses.

School Name	# of Geology Courses Offered	# of Geoinformatics Courses Offered	%
University of Minnesota - Twin Cities	59	1	1.7
University of Texas - Austin	81	2	2.5
Standford University (lowest % in the West-USA)	34	1	2.9
Pennslyvannia State University - University Park	83	3	3.6
Yale University	53	2	3.8
Indiana University - Bloomington	46	2	4.3
University of Colorado - Boulder	65	3	4.6
Rice University	70	4	5.7
University of California - Santa Barbara	86	5	5.8
University of California - Los Angeles	60	4	6.7
Massachusetts Institute of Technology	42	3	7.1
Colorado School of Mines	41	3	7.3
California Institute of Technology	41	3	7.3
Ohio State University	67	5	7.5
University of Wisconsin - Madison	51	4	7.8
University of California - Berkeley	36	3	8.3
University of Washington	42	4	9.5
Columbia University (highest % in the East-USA)	30	4	13.3

[% 0-4 red; 5-6 yellow; 7+ green]

- Although some courses teach skills and concepts useful in geoinformatics application, they are not considered primarily a geoinformatics course because their focus is still on understanding geological phenomena and processes
- GIS map (Fig. 1) was produced in R programming language using leaflet, sf, and leaflet.minichart packages (Cheng et al., 2022)

Results

- Top-ranked colleges in geology programs had few geoinformatics courses offered, with the maximum being five courses
- Western colleges offered more geoinformatics courses than Eastern colleges. The proportion of geoinformatics courses offered relative to the total number of geology courses offered exhibited a similar pattern
- Exceptions to the pattern: Columbia University (highest) and Stanford University (lowest)

Discussion

- Differences in cultural attitudes and values between the West and the East can explain the phenomenon
- The settlement history of the regions and the legacy of different industries and social movements within the regions may contribute to this.
- West: has history of being a technological hub, with Silicon Valley as a prime example; the presence of leading technology companies and access to venture capital → tendency to be more entrepreneurial, innovative, and forward-thinking
- East: has a longer history of settlement, established traditions, and cultural norms; home to many of the country's oldest and most established cultural institutions more cautious and conservative; industries are more traditional, such as finance, law, and manufacturing → tendency to be more conservative and cautious for maintaining stability

Conclusion

- ❑ The Western high-ranked US colleges provide more geoinformatics courses than the Eastern with few exceptions
- ❑ The West has a stronger history and cultural attitude towards advanced technology and innovation, as evidenced by Silicon Valley.
- ❑ Geoinformatics is important in several geo-applications and disciplines, and the U.S. education system should prioritize its inclusion in earth-science education, necessitating a standardized curriculum and further research in the area.

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